

CS4201: Information Retrieval and Web Search

Introduction to the course

Dwaipayan Roy

Department of Computational and Data Sciences

Course overview

To study:

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- How do search engines work?

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- Why do search engines fail?
- How can they be evaluated?

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- We need smarter tools for efficient searching and other processing of these data.
- IR provides tools for analyzing and organizing content to facilitate search, discovery, and learning.

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- **Lucene.**

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- **PyLucene**.

What CS4201 is **not** about

- Machine Learning and Deep Learning.

What CS4201 is not about

- Machine Learning and Deep Learning. but...

Regular schedule

Tuesday	8:00AM - 9:50PM
Wednesday	10:45AM - 11:40PM
Friday	1:30PM - 2:25PM

Academic calendar - Spring 2026

CS4201 classes

January	4 weeks	Introduction, Text processing Text processing, Lemmatization, Stemming, Boolean Retrieval Index creation, Compression, Statistical properties of text Lucene - 1, Query representation and processing	Assignment/Class test 1
February	3 weeks	Ranked retrieval, Term weighting VSM Probabilistic retrieval, BM25 Language modeling for information retrieval	Mid sem
March	3 weeks	Lucene - 2 Retrieval system evaluation Relevance feedback, Rocchio, Relevance based language model	Assignment/Class test 2
April	2 weeks	Web search, Link analysis Revise	End sem

- 14.02.26 - 21.02.26 – Mid-semester exam
- 01.03.26 - 08.03.26 – Spring break
- 17.04.26 – Last day for conduct of classes

- 23.01.26 – Saraswati Puja
- 31.03.26 – Mahavir Jayanti
- 03.04.26 – Good Friday

Interaction

dwaipayan.roy@iiserkol.ac.in

- Send email to cs4201@iiserkol.ac.in for interactions concerning all participants.

Course grading (tentative)

30%	mid-sem exam
50%	end-sem exam
20%	assignment/class test

Lecture format

- Approx 45 minutes per class.
 - ▶ will start 5 minutes after the scheduled start time;
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- Feel free to ask any question (how trivial that may sound to yourself!).

References

- (MRS) *Introduction to Information Retrieval*
C. Manning, P. Raghavan and H. Schutze
<https://nlp.stanford.edu/IR-book/information-retrieval-book.html>
- (BCC) *Information Retrieval: Implementing and Evaluating Search Engines*
S. Buttcher, C. L. A. Clarke, G. Cormack.
<http://www.ir.uwaterloo.ca/book/>
E-copy: <https://tinyurl.com/IRBook-Buttcher>

Information Retrieval



www.trec.nist.gov

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- **Information Retrieval:** process of seeking out **documents** of *unstructured nature*, that satisfies the *information need* (of a user) from within a **collection of documents**.
- Document types:
 - ▶ **text**, audio, video, images etc.

Information Retrieval: Text Document

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 - ▶ *Unstructured nature*: No specific structure (just a collection of words).

Information Retrieval: Text Document

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 - ▶ *Unstructured nature*: No specific structure (just a collection of words).
 - ▶ *Information need*: The query of the user.

Information Retrieval: The Root



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Library science, now often called Library and Information Science (LIS), is an interdisciplinary field focused on managing, organizing, preserving, and providing access to information resources in all formats (print, digital) for people, merging traditional library practices with modern technology and information management.



Information Retrieval: A Practical Application



www.google.com

Information Retrieval: A Graphical Representation

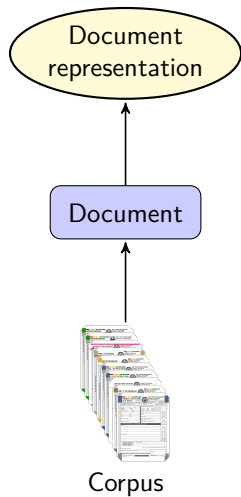


Corpus

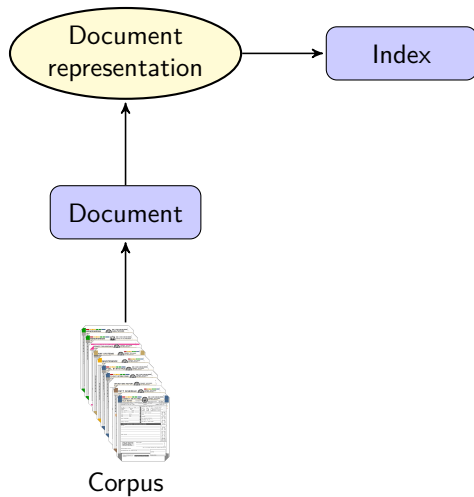
Information Retrieval: A Graphical Representation



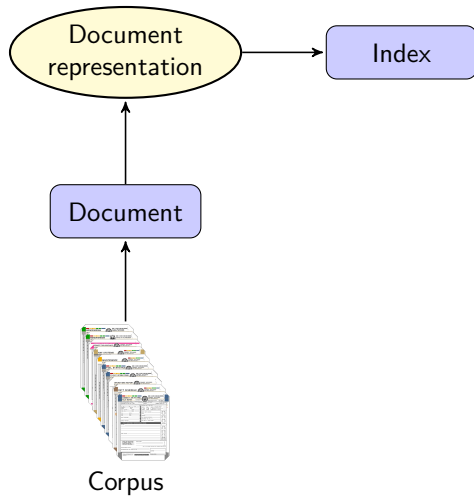
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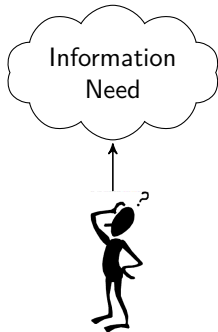
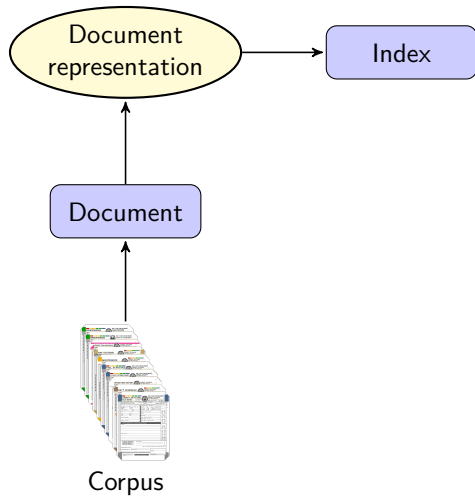
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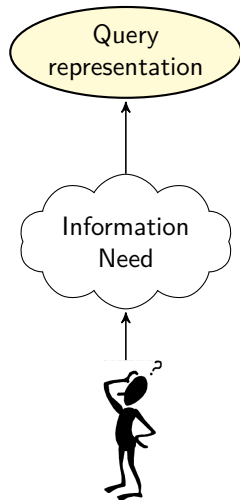
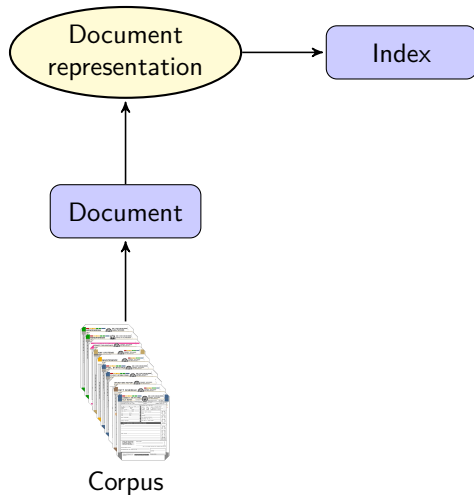
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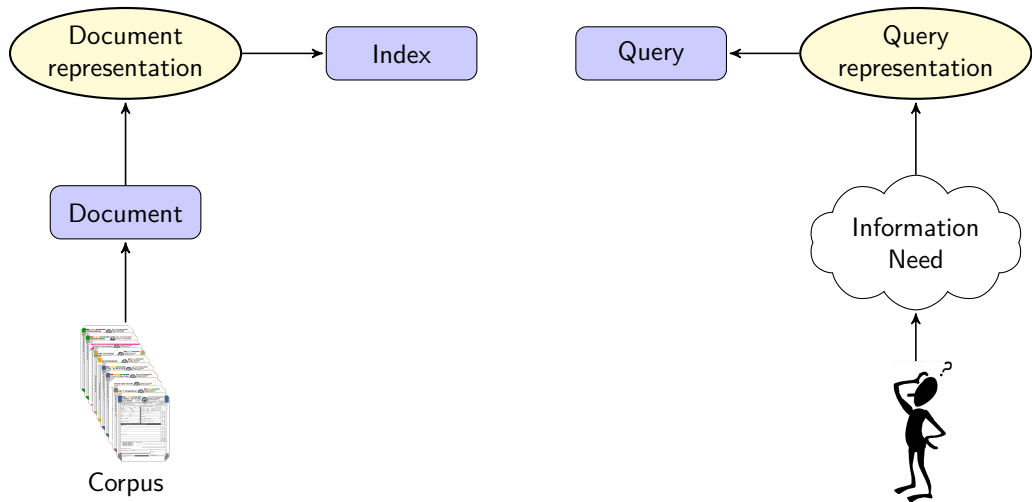
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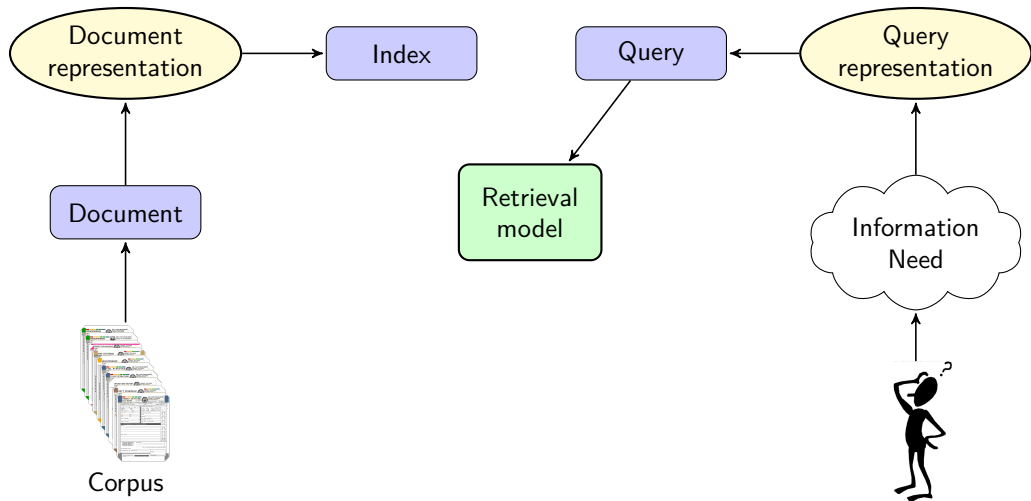
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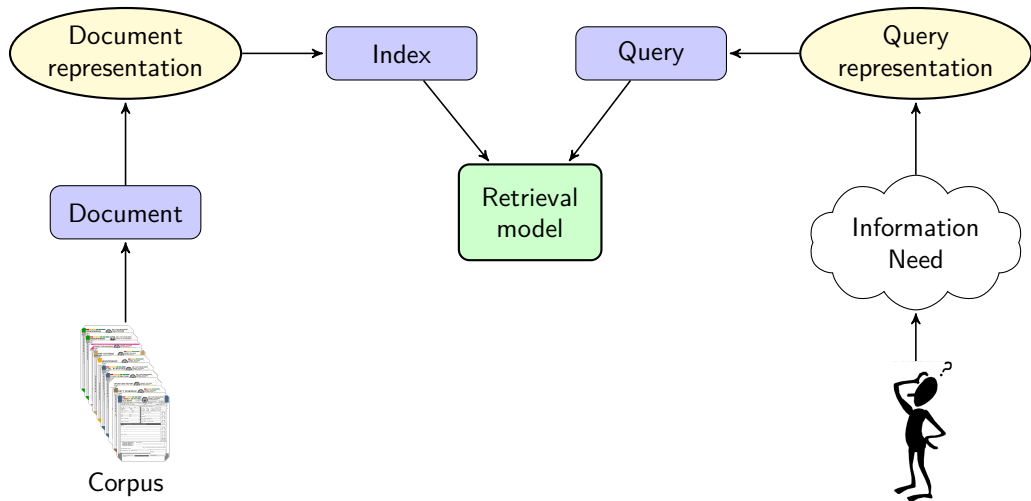
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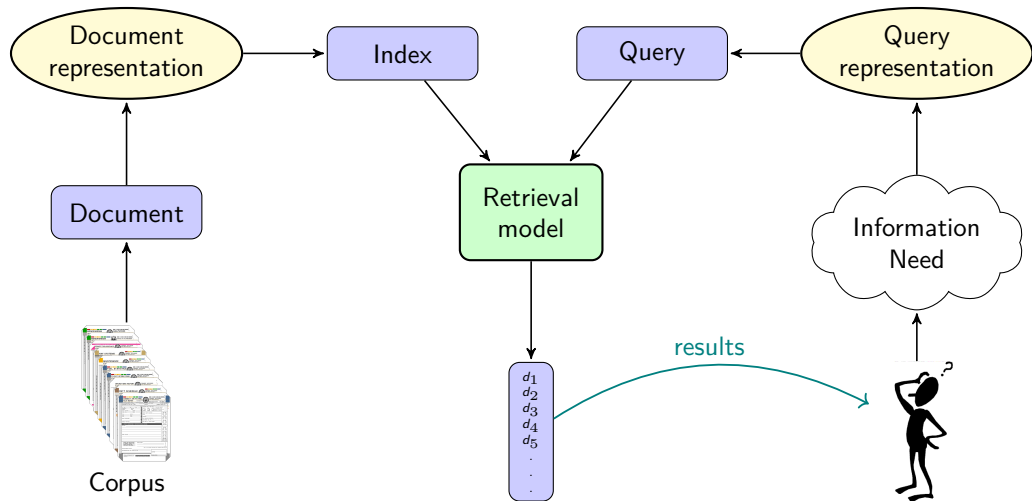
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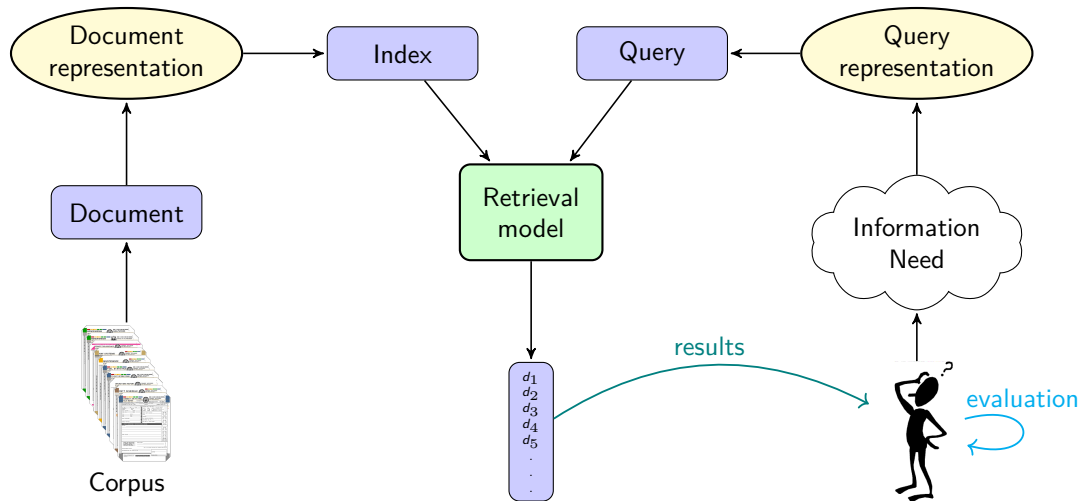
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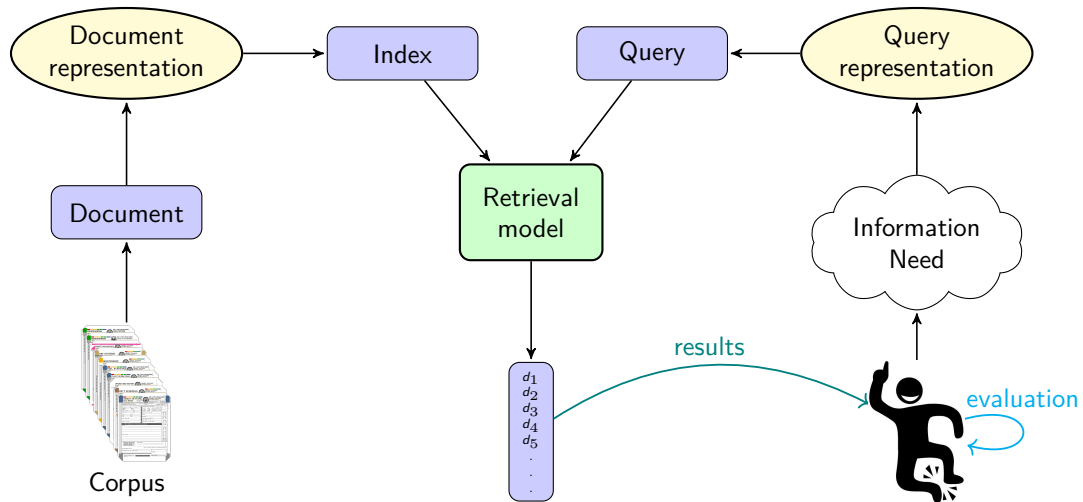
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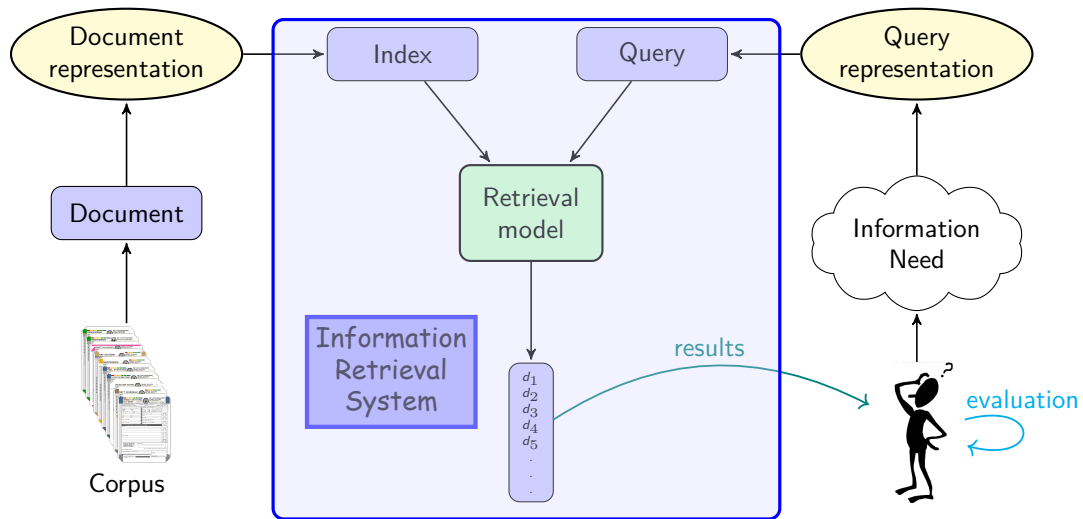
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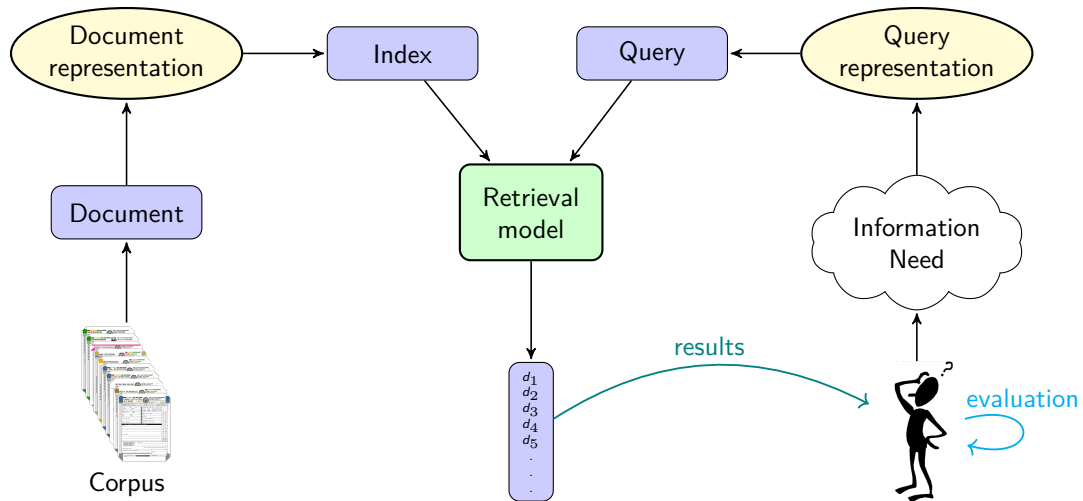
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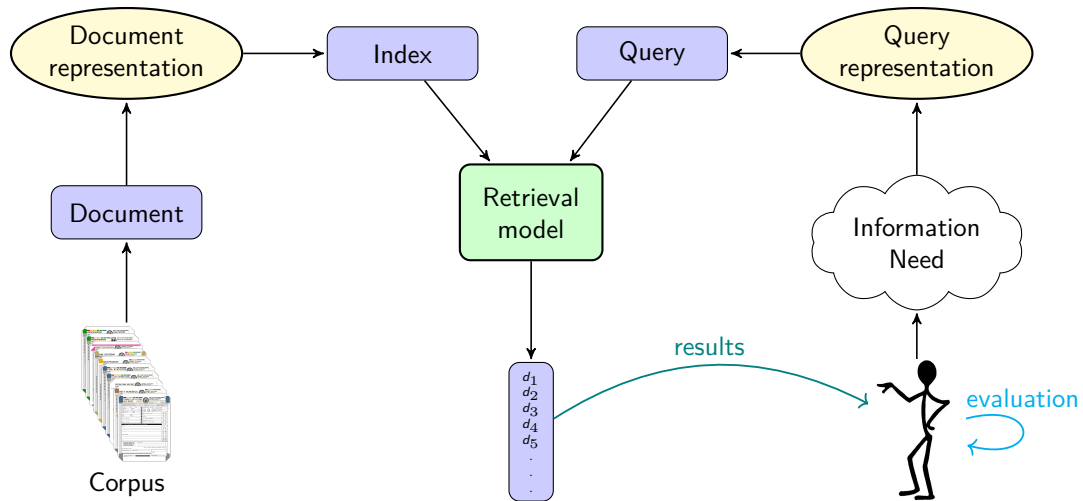
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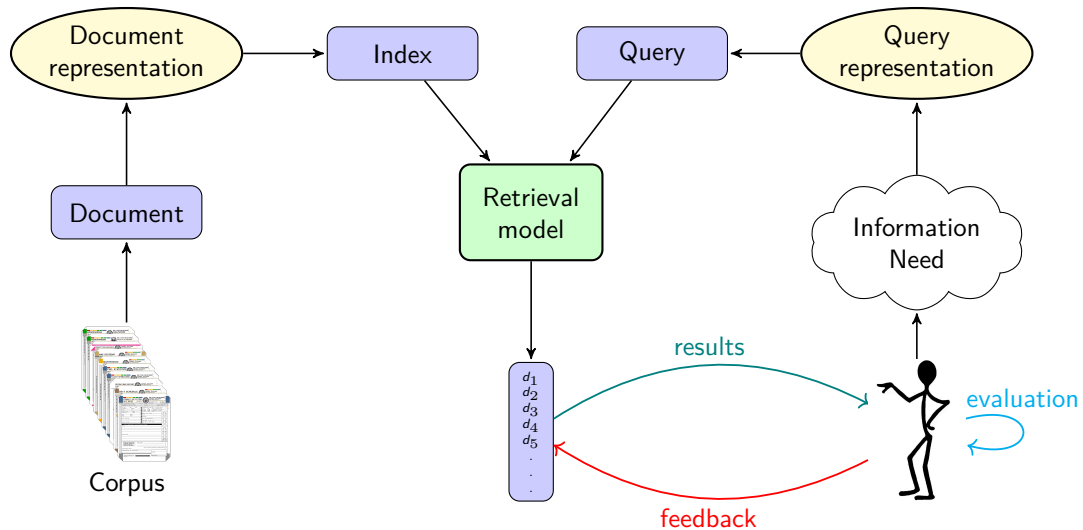
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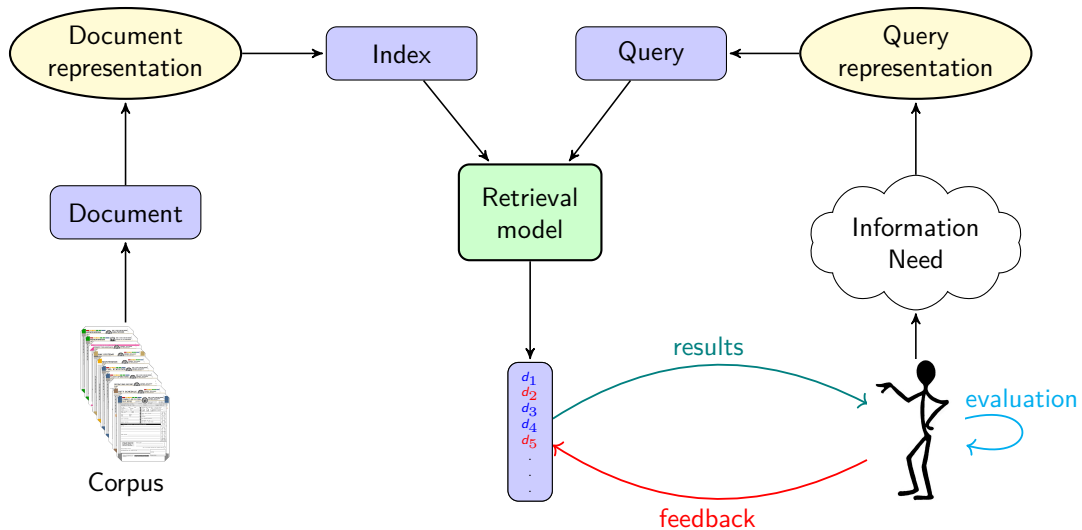
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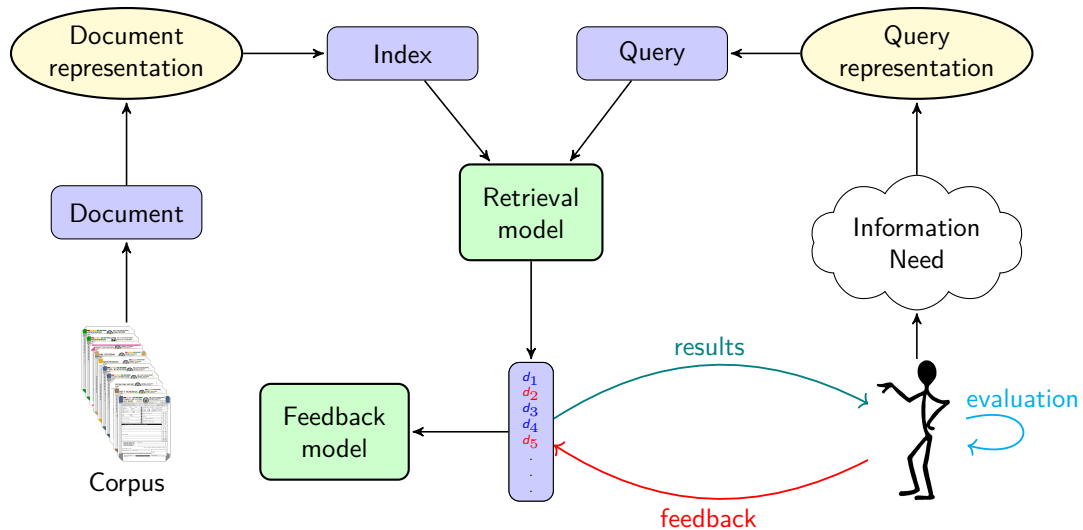
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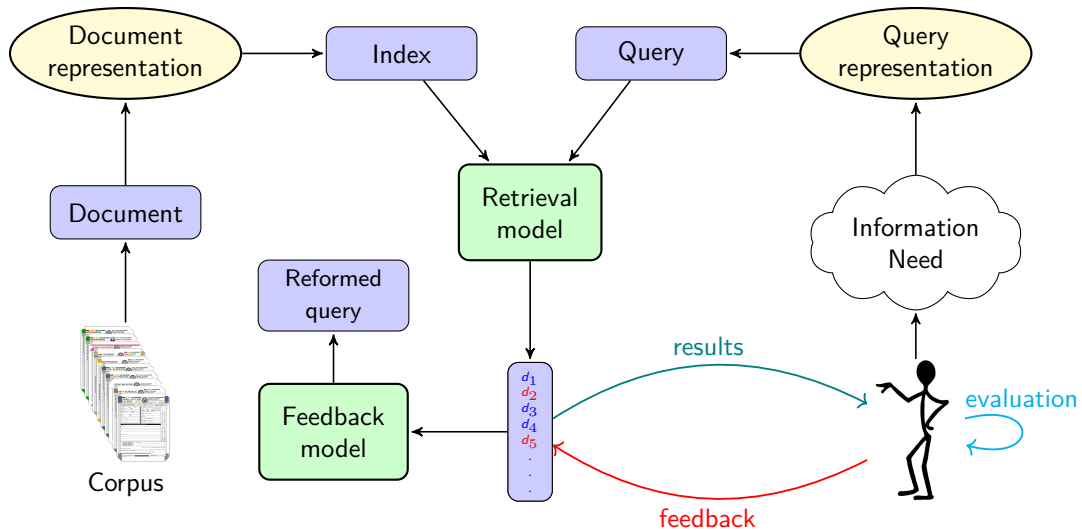
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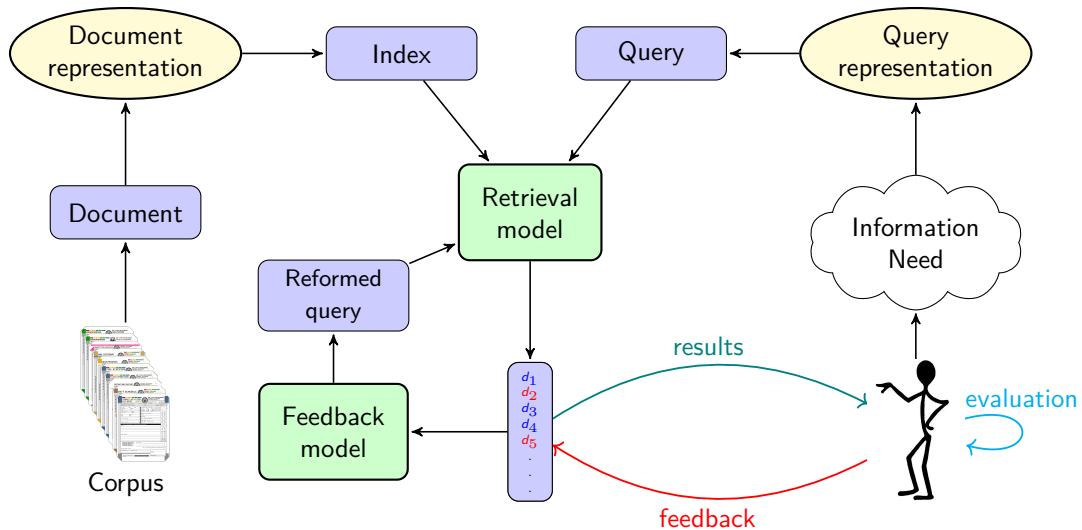
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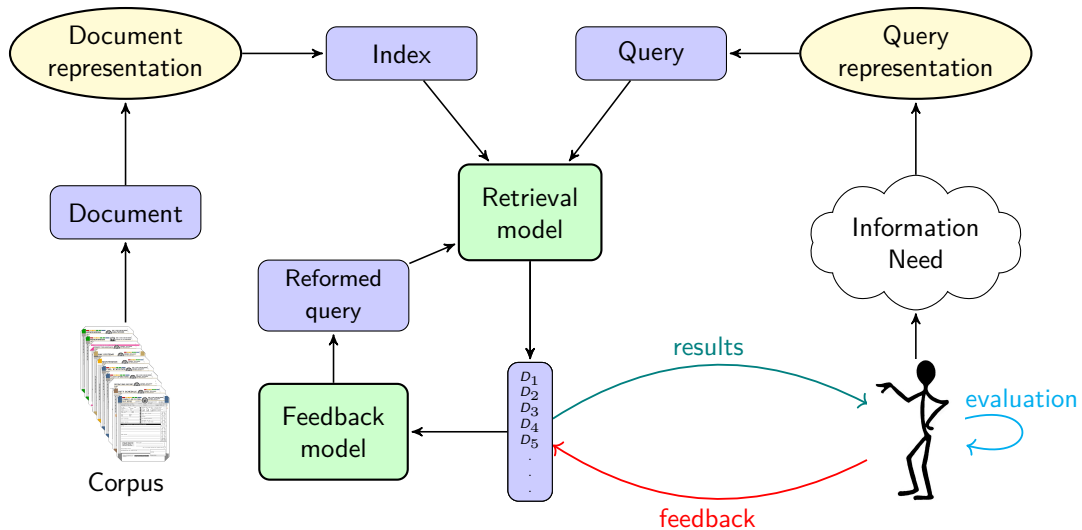
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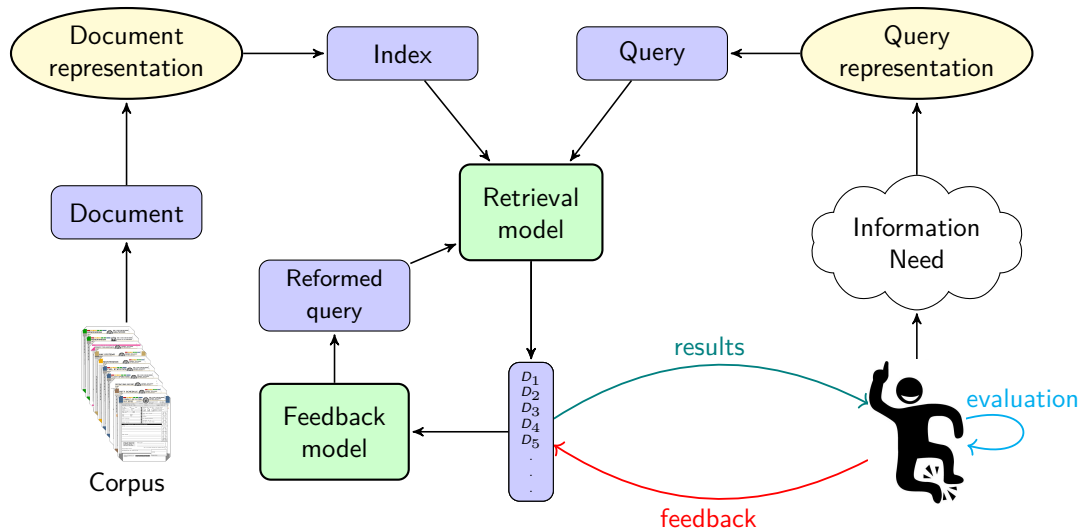
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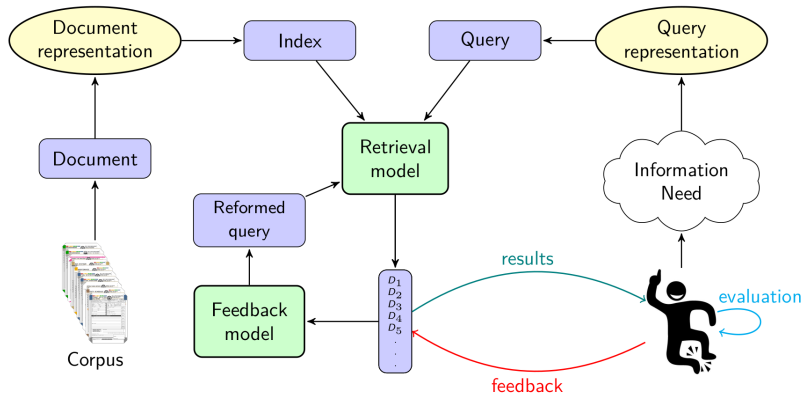
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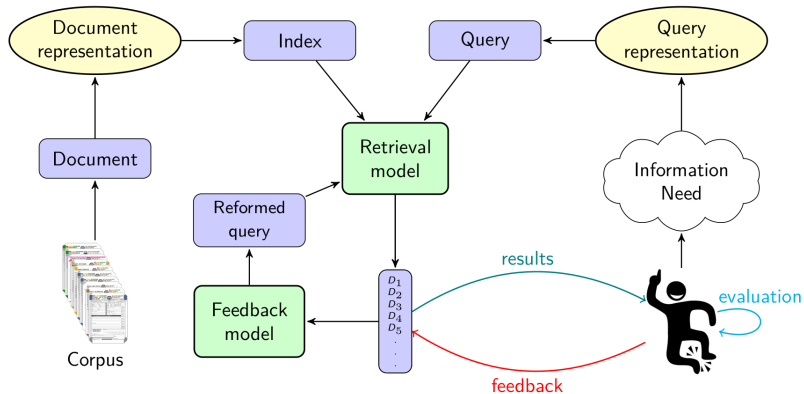


CS4201 - topics to be covered



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- Query representation
- Retrieval model
- Result evaluation
- Feedback model

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